

# Bounds for hypergraph Universality at D=1

Sifei Li

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For any log without indicating its base, we assume it is base on 2.

## 1 Prerequisites

**Theorem 1** (Chernoff bound). *Let  $X = \sum_{i=1}^n X_i$ , where the  $X_i$  are independent Bernoulli random variables. Then*

1.  $\mathbb{P}(X \geq (1 + \delta)\mathbb{E}X) \leq \exp\left(-\frac{\delta^2}{2+\delta}\mathbb{E}X\right)$  for all  $\delta > 0$ ,
2.  $\mathbb{P}(X \leq (1 - \delta)\mathbb{E}X) \leq \exp\left(-\frac{\delta^2}{2}\mathbb{E}X\right)$  for all  $0 < \delta < 1$ .

The observation 2.1 in the paper [Kat25] still hold for  $D = 1$ , which is a 1-degenerate  $r$ -graph has at most  $rn/k$  vertices of degree at least  $k$ .

## 2 Proofs

**Definition 1** (The block model for the case  $D = 1$ ). *Fix  $r \geq 3$ , and let  $q := r - 1$ . Let  $N$  be the smallest integer such that*

$$n^{q^{1-N}} \leq 3^{q^2}.$$

For  $1 \leq k \leq N$ , define

$$\Delta_k := n^{q^{-k}},$$

and set  $\Delta_0 := n$ , so  $\Delta_{k-1} = \Delta_k^q$ . Let  $W = W_1 \sqcup \cdots \sqcup W_N$  be a disjoint union of blocks satisfying

$$|W_k| = \frac{100 \cdot 3^{qr} n}{\Delta_k} \quad \text{for each } 1 \leq k \leq N.$$

Each block  $W_k$  is further partitioned into sub-blocks

$$W_k = W_{k,1} \sqcup \cdots \sqcup W_{k,\log n},$$

where

$$|W_{k,1}| = \frac{1}{2}|W_k| \quad \text{and} \quad |W_{k,j}| \geq \frac{|W_k|}{2 \log n} \quad \text{for all } j \geq 2.$$

For  $s \in \binom{W}{r}$ , let  $\pi_i(s) := |s \cap W_i|$ , and define

$$p^* := K_r \frac{(\log n)^2}{n}, \quad p_s := \min \left\{ 1, p^* \prod_{i=1}^N \Delta_i^{\pi_i(s)} \right\},$$

where  $K_r$  is a constant depending only on  $r$ . The formula of  $K_r$  comes from the proof of Lemma 2. The random  $r$ -graph  $\Gamma$  on vertex set  $W$  is obtained by including each  $s \in \binom{W}{r}$  as its edge independently with probability  $p_s$ .

**Proposition 1.** Fix  $r \geq 3$ , for all sufficiently large  $n$ , all the definitions follow Definition 1. The followings hold: (may insert more if needed)

1.  $N < \log \log n$

*Proof.* from the definition of  $N$  in Definition 1, we have  $n^{q^{1-(N-1)}} > 3^{q^2}$ . Taking  $\log_3$  and  $\log_n$  on both sides gives  $(2 - N) + \log_q \log_3 n > 2$ . Rearranging the equation gives:

$$N < \log_q \log_3 n \leq \log \log n$$

as desired.  $\square$

**Proposition 2** (The  $D = 1$  version of Proposition 4.2(c)). For every fixed  $r \geq 3$ , there is a constant  $C_r > 0$  such that, for all sufficiently large  $n$ , the random  $r$ -graph  $\Gamma = \Gamma(r, n, 1)$  defined in Definition 1 has at most  $150 \cdot 3^q r n = 150 \cdot 3^{r-1} r n$  vertices, and asymptotically almost surely has at most

$$C_r (\log \log n)^r (\log n)^2 n^{r-1}$$

edges with some constant  $C_r$  depending on  $r$ .

*Proof.* We first bound the number of vertices. By the definition of  $N$ , we have

$$\Delta_{N-1} = n^{q^{1-N}} \leq 3^{q^2} \text{ and } \Delta_{N-2} = n^{q^{2-N}} > 3^{q^2}.$$

Raise the left equation to the power  $1/q$  and combine with the right equation,

$$3^q \geq \Delta_{N-2}^{1/q^2} = \Delta_N > 3.$$

Moreover, using the property  $\Delta_{k-1} = \Delta_k^q$  for  $0 \leq k \leq N$ , and  $q = r - 1 \geq 2$ , we get

$$\begin{aligned} |W| &= |W_1| + |W_2| + \dots + |W_N| = 100 \cdot 3^q r n \sum_{k=1}^N \frac{1}{\Delta_k} = 100 \cdot 3^q r n \sum_{s=0}^{N-1} \frac{1}{\Delta_N^{q^s}} \\ &\leq 100 \cdot 3^q r n \sum_{s=0}^{\infty} \frac{1}{\Delta_N^{q^s}} \leq 100 \cdot 3^q r n \sum_{s=0}^{\infty} \frac{1}{3^{2^s}} = 100 \cdot 3^q r n \frac{1}{1 - \frac{1}{3}} = 150 \cdot 3^q r n. \end{aligned}$$

Thus  $\Gamma$  has at most  $150 \cdot 3^q r n$  vertices.

We now estimate the expected number of edges. Fix an ordered  $r$ -tuple

$$S = (i_1, \dots, i_r) \in [N]^r.$$

For each  $k \in [N]$ , let  $m_S(k)$  be the multiplicity of  $k$  in  $S$ . The expected number of edges of  $\Gamma$  with one vertex in each of  $W_{i_1}, \dots, W_{i_r}$  is at most

$$\min \left\{ 1, p^* \prod_{\ell=1}^r \Delta_{i_\ell} \right\} \prod_{k \in [N]} |W_k|^{m_S(k)} \leq p^* \prod_{\ell=1}^r \Delta_{i_\ell} \prod_{k \in [N]} |W_k|^{m_S(k)}.$$

Thus

$$\mathbb{E}[|E(W_{i_1}, \dots, W_{i_r})|] \leq p^* \prod_{\ell=1}^r \Delta_{i_\ell} \prod_{k \in [N]} |W_k|^{m_S(k)} = p^* \prod_{\ell=1}^r \Delta_{i_\ell} \frac{(100 \cdot 3^q r n)^2}{\Delta_{i_\ell}} = p^* \prod_{\ell=1}^r 100 \cdot 3^q r n$$

Summing over all  $N^r$  choices of  $r$ -tuples and using  $N \leq \log \log n$  from Proposition 1, we get

$$\mathbb{E}|E(\Gamma)| \leq N^r p^* \prod_{\ell=1}^r 100 \cdot 3^q r n = N^r p^* (100 \cdot 3^q r n)^r \leq (\log \log n)^r \frac{K_r (\log n)^2}{n} (100 \cdot 3^q r n)^r := C'_r (\log \log n)^r (\log n)^2 n^{r-1},$$

where  $C'_r = K_r(100 \cdot 3^q \cdot r)^r$ .

It remains to turn this expectation bound into an asymptotically almost sure bound. Denote  $M = C'_r(\log \log n)^r(\log n)^{2n^{r-1}}$  and apply part 1 of Theorem 1 (Chernoff bound) with

$$\delta = 2 \frac{M}{\mathbb{E}|E(\Gamma)|} - 1,$$

notice that  $\delta \geq 1$ , we get

$$\begin{aligned} & \mathbb{P}[|E(\Gamma)| \geq 2M] \\ &= \mathbb{P}(|E(\Gamma)| \geq (1 + \delta)\mathbb{E}|E(\Gamma)|) \\ &\leq \exp\left[-\frac{\delta^2}{2 + \delta}\mathbb{E}|E(\Gamma)|\right] \\ &\leq \exp\left[-\frac{\delta\mathbb{E}|E(\Gamma)|}{3}\right] \\ &\leq \exp[-M/3] \\ &= o(1) \end{aligned}$$

Taking  $C_r = 2C'_r = 2K_r(100 \cdot 3^q \cdot r)^r$  finishes the proof.  $\square$

**Definition 2** (Embedding Strategy). *[Kat25] Given any  $D$ -degenerate  $r$ -graph  $H$ , let  $v_1, \dots, v_n$  be a  $D$  degeneracy ordering of  $V(H)$ , and let  $\Gamma$  have the properties of Proposition 2. Our embedding strategy is defined by the iterative construction of a partial embedding  $\psi_t : \{v_1, \dots, v_t\} \rightarrow \Gamma$ . We start with  $\psi_0$ , the trivial partial embedding of no vertices into  $\Gamma$ .*

*For each  $1 \leq t \leq n$ , we say a vertex  $u \in \Gamma$  is a candidate for  $v_t$  if  $\psi_{t-1}(L^-(v_t)) \subseteq L(u)$ . We say such a vertex  $u \in \Gamma$  is an available candidate if in addition  $u \notin \text{Im}(\psi_{t-1})$ . Let  $k$  be minimal such that  $\Delta_k < \deg(v_t)$ . If there is no such  $k$  choose  $k = N$ . Choose  $j$  minimal such that there is some  $u \in W_{k,j}$  which is an available candidate for  $v_t$ . We define  $\psi_t = \psi_{t-1} \cup \{v_t \rightarrow u\}$ . If there is no available candidate for  $v_t$  in  $W_k$ , we say  $\psi_t$  and the subsequent partial embeddings do not exist and that the embedding strategy fails at time  $t$ . If the embedding strategy does not fail, then  $\psi := \psi_n$  gives an embedding of  $H$  into  $\Gamma$ .*

This embedding strategy will not change for the case  $D = 1$ , same as the definition of well-behaved multisets.

**Definition 3** (Well-behaved multisets for the case  $D = 1$ ). *Let  $\mathcal{B} = (B_1, \dots, B_t)$  be a multiset such that each  $B_i \subseteq \binom{W}{r-1}$ . For  $B_i \subseteq \binom{W}{r-1}$ , write*

$$V(B_i) := \bigcup_{b \in B_i} b,$$

*and write*

$$V(\mathcal{B}) := \bigcup_{i=1}^t V(B_i).$$

*We say that  $\mathcal{B}$  is well-behaved if the following three conditions hold:*

$$|B_i| \leq 1 \quad \text{for all } i \in [t], \tag{WB1}$$

$$|\{i \in [t] : u \in V(B_i)\}| \leq \Delta_{k-1} \quad \text{for all } 1 \leq k \leq N \text{ and all } u \in W_k, \tag{WB2}$$

*and*

$$|V(\mathcal{B}) \cap W_{k,j}| \leq \frac{1}{2}|W_{k,j}| \quad \text{for all } 1 \leq k \leq N \text{ and all } 1 \leq j \leq \log n. \tag{WB3}$$

**Lemma 1.** *Let  $r \geq 3$ , let  $n$  be sufficiently large, and let  $t \in [n]$ . Let  $\Gamma$  be the random block model defined in Definition 1, and let  $\mathcal{B} = (B_1, \dots, B_t)$  be a well-behaved multiset of size  $t$ . Fix  $1 \leq k \leq N$  and  $v \in W_k \setminus V(\mathcal{B})$ . Let  $\mathcal{E}_v$  be the event that there exists some  $B_i \in \mathcal{B}$  such that  $B_i \subseteq L(v)$ . Then*

$$\mathbb{P}(\mathcal{E}_v) \geq \min \left\{ \frac{1}{2}, \frac{1}{2} tp^* \Delta_k \right\}.$$

*Proof.* Let  $1 \leq k \leq N$ , and  $v \in W_k \setminus V(\mathcal{B})$ . If  $B_i = \emptyset$  for some  $i \in [t]$ , then  $B_i \subseteq L(v)$  always holds, and hence  $\mathbb{P}(\mathcal{E}_v) = 1$ . We may therefore assume that  $B_i \neq \emptyset$  for every  $i \in [t]$ . By (WB1), for each  $i \in [t]$  there is an  $(r-1)$ -set  $b_i \in \binom{W}{r-1}$  such that  $B_i = \{b_i\}$ .

Let  $S := \{b_i : i \in [t]\}$  be the set of distinct  $(r-1)$ -sets appearing among the  $b_i$ 's, and for  $b \in S$ , let  $m_b := |\{i \in [t] : b_i = b\}|$  be the multiplicity of  $b$ , and we have  $\sum_{b \in S} m_b = t$ .

Fix  $b \in S$ , and write  $b = \{x_1, \dots, x_{r-1}\}$ , where  $x_l \in W_{k_l}$  for  $1 \leq l \leq r-1$ . Condition (WB2) gives

$$m_b \leq |\{i \in [t] : x_l \in V(B_i)\}| \leq \Delta_{k_l-1} \quad \text{for each } 1 \leq l \leq r-1.$$

For each  $1 \leq k_l \leq N$ ,  $\Delta_{k_l} = \Delta_{k_l-1}^{1/q} \geq m_b^{1/q}$ . Therefore

$$\prod_{l=1}^{r-1} \Delta_{k_l} \geq m_b^{(r-1)/q} = m_b.$$

Let  $E_b$  be the event that  $b \in L(v)$ . Since  $v \notin V(\mathcal{B})$ , the set  $\{v\} \cup b$  has size  $r$ , and  $E_b$  is exactly the event that  $\{v\} \cup b$  is an edge of  $\Gamma$ . Therefore

$$\mathbb{P}(E_b) = p_{\{v\} \cup b} = \min \left\{ 1, p^* \Delta_k \prod_{\ell=1}^q \Delta_{k_\ell} \right\} \geq \min\{1, p^* \Delta_k m_b\}.$$

The events  $(E_b)_{b \in S}$  are independent, since they concern distinct  $r$ -sets of the form  $\{v\} \cup b$ . Thus,  $\mathcal{E}_v = \bigcup_{b \in S} E_b$ , and  $\mathbb{P}(\mathcal{E}_v) = 1 - \prod_{b \in S} (1 - \mathbb{P}(E_b))$ .

Furthermore, notice that

$$\sum_{b \in S} \mathbb{P}(E_b) \geq \sum_{b \in S} \min\{1, p^* \Delta_k m_b\} \geq \min \left\{ 1, p^* \Delta_k \sum_{b \in S} m_b \right\} = \min\{1, tp^* \Delta_k\}.$$

Since  $1 - x \leq e^{-x}$  for all  $x \in [0, 1]$  (which follows from the Taylor expansion  $e^{-x} = \sum_{k=0}^{\infty} \frac{(-x)^k}{k!} = 1 - x + \frac{x^2}{2} - \dots \geq 1 - x$ ), we have

$$\mathbb{P}(\mathcal{E}_v) \geq 1 - \exp \left( - \sum_{b \in S} \mathbb{P}(E_b) \right) \geq 1 - \exp(-\min\{1, tp^* \Delta_k\}).$$

Finally, notice that for every  $x \geq 0$ ,  $1 - e^{-\min\{1, x\}} \geq \frac{1}{2} \min\{1, x\}$ , which can be verified by considering two cases: If  $x \leq 1$ , then  $1 - e^{-x} \geq x - \frac{x^2}{2} \geq \frac{x}{2}$  by Taylor expansion again; if  $x \geq 1$ , then  $1 - e^{-1} > \frac{1}{2}$ . Applying this with  $x = tp^* \Delta_k$ , we get

$$\mathbb{P}(\mathcal{E}_v) \geq \frac{1}{2} \min\{1, tp^* \Delta_k\} = \min \left\{ \frac{1}{2}, \frac{1}{2} tp^* \Delta_k \right\},$$

as required.  $\square$

**Lemma 2.** *Let  $r \geq 3$  and  $n$  be sufficiently large. Then  $\Gamma$  defined in the Definition 1 a.a.s. satisfies the following:  $\forall t \leq n$  and well-behaved multiset  $\mathcal{B}$  of size  $t$ .  $\forall k \leq N, j \leq \log n$ , we have*

$$|\{u \in W_{k,j} : \exists B_i \in \mathcal{B}, B_i \subseteq L(u)\}| \geq \min \left\{ \frac{1}{16}, \frac{tp^* \Delta_k}{8} \right\} |W_{k,j}| \quad (1)$$

*Proof.* Let  $r, n, k, j$  be integers.  $r \geq 3$ ,  $n$  large enough. Let  $t \leq n$  and  $\mathcal{B}$  be a well-behaved multiset of size  $t$ . Fix  $1 \leq k \leq N$  and  $1 \leq j \leq \log n$ . If  $tp^*\Delta_k > 1$ , then we can pick a  $t'$  such that  $\frac{1}{2} < t'p^*\Delta_k \leq 1$ , and arbitrarily choose  $\mathcal{B}' \subseteq \mathcal{B}$  with  $|\mathcal{B}'| = t'$ . Then

$$|\{u \in W_{k,j} : \exists B_i \in \mathcal{B}, B_i \subseteq L(u)\}| \geq \frac{1}{16}|W_{k,j}|$$

as desired. Notice that such  $t'$  exists as  $p^*\Delta_k$  is small enough, indeed,  $p^*\Delta_k \leq K_r(\log n)^2 n^{(1/q)-1} = o(1)$  with  $q \geq 2$ . Thus we first prove the lemma under the assumption  $tp^*\Delta_k \leq 1$ , then the large case follows by applying the already-proved small case to the sub-multiset  $\mathcal{B}'$ .

If there exists an  $i$  such that  $B_i = \emptyset$ , then  $B_i \subseteq L(u)$  for every  $u \in W_{k,j}$ , and the desired inequality is immediate.

Let  $U := W_{k,j} \setminus V(\mathcal{B})$ . By (WB3),  $|U| \geq \frac{1}{2}|W_{k,j}|$ . For each  $u \in U$ , let  $X_u$  be the indicator of the event  $\mathcal{E}_u$  that there exists  $i \in [t]$  such that  $B_i \subseteq L(u)$ , and set  $X := \sum_{u \in U} X_u$ .

The random variables  $(X_u)_{u \in U}$  are independent. Indeed,  $X_u$  depends only on edge indicators of sets of the form  $\{u\} \cup b$ , where  $b$  is the unique element of some  $B_i$ . By definition, the edge indicator is an independent random variable, so  $X_u$  is also an independent random variable.

By Lemma 1, for every  $u \in U$ ,  $\mathbb{P}(\mathcal{E}_u) \geq \min\{\frac{1}{2}, \frac{1}{2}tp^*\Delta_k\}$ . Since  $tp^*\Delta_k \leq 1$ , this gives  $\mathbb{P}(\mathcal{E}_u) \geq tp^*\Delta_k/2$ . Therefore,

$$\mathbb{E}X = \sum_{u \in U} \mathbb{E}X_u \geq \frac{1}{2}tp^*\Delta_k|U| \geq \frac{1}{4}tp^*\Delta_k|W_{k,j}| := 2T. \quad (2)$$

where  $T = \frac{1}{8}tp^*\Delta_k|W_{k,j}|$ .

Now we show that the probability that the conclusion fails is  $o(1)$ . By part 2 of Theorem 1,  $\Pr(X \leq T) \leq \Pr(X \leq (1 - \frac{1}{2})\mathbb{E}X) \leq \exp(-\frac{1}{8}\mathbb{E}X)$ . To obtain a lower bound on  $\mathbb{E}X$ , note that  $|W_{k,j}| \geq \frac{|W_k|}{2 \log n} = \frac{50 \cdot 3^q r n}{\Delta_k \log n}$ , therefore

$$\mathbb{E}(X) \geq \frac{1}{4}tp^*\Delta_k|W_{k,j}| \geq \frac{1}{4}tp^*\Delta_k \frac{50 \cdot 3^q r n}{\Delta_k \log n} \geq 25/2 \cdot 3^q r K_r t \log n.$$

Combining all together and using  $\log n = \frac{\ln n}{\ln 2}$ ,

$$\Pr(X < T) \leq \exp\left(-\frac{1}{8} \cdot 25/2 \cdot 3^q r K_r t \log n\right) = n^{-25 \cdot 3^q \cdot r \cdot K_r \cdot t / (16 \ln 2)} = n^{-\alpha t}$$

where  $\alpha = 25 \cdot 3^q \cdot K_r / (16 \ln 2)$ . Finally, we count the number of choices. By Proposition 2,  $|V(\Gamma)| \leq 150 \cdot 3^q r n$ . Thus each  $B_i$  has at most  $\binom{|V(\Gamma)|}{r-1} + 1 \leq (150 \cdot 3^q r)^r n^r$  possible choices.

Therefore the number of well-behaved multisets  $\mathcal{B}$  of size  $t$  is at most  $(150 \cdot 3^q r)^{rt} n^{rt}$ . By Proposition 1, there are at most  $N \leq \log \log n$  choices for  $k$ ,  $\log n$  choices for  $j$ .

Hence the probability that Lemma 2 fails for some fixed  $t$  is at most

$$(\log \log n)(\log n) (150 \cdot 3^q r)^{rt} n^{rt} n^{-\alpha t}.$$

Recall that we picked  $K_r$  sufficiently large so that

$$r(1 - \alpha) \leq -5,$$

then

$$(\log \log n)(\log n) (150 \cdot 3^q r)^{rt} n^{rt} n^{-\alpha t} \leq (150 \cdot 3^q r)^{rt} (\log \log n)(\log n) n^{-5t} \leq (\log \log n)(\log n) n^{-4t}$$

using the fact that  $(150 \cdot 3^q r)^r \leq n$ , as  $n$  is sufficiently large.

Finally, taking a union bound over all  $t \leq n$ ,

$$\Pr(\text{Lemma 2 fails}) \leq \sum_{t=1}^n (\log \log n)(\log n) n^{-4t} \leq (\log \log n)(\log n) n^{-4} \cdot n \leq n^{-1} = o(1).$$

This completes the proof.  $\square$

**Theorem 2.** For  $r \geq 3$ , there exists a constant  $C_r$  such that for all sufficiently large  $n$ , there exists an  $r$ -uniform hypergraph  $\Gamma$  with

$$e(\Gamma) \leq C_r n^{r-1} (\log n)^2 (\log \log n)^r$$

which is universal for the class  $\mathcal{H}^{(r)}(n, 1)$  of  $n$ -vertex 1-degenerate  $r$ -graphs.

*Proof.* Let  $r \geq 3$ . By Proposition 2, the random  $r$ -graph  $\Gamma$  defined in Definition 1 asymptotically almost surely satisfies

$$e(\Gamma) \leq C_r n^{r-1} (\log n)^2 (\log \log n)^r,$$

and the statement of Lemma 2. It therefore remains to show that  $\Gamma$  is  $\mathcal{H}^{(r)}(n, 1)$ -universal.

Let  $H \in \mathcal{H}^{(r)}(n, 1)$ . Since  $H$  is 1-degenerate, there exists an ordering  $v_1, \dots, v_n$  of  $V(H)$  such that every vertex  $v_t$  has at most one back-link. We embed  $H$  into  $\Gamma$  using the embedding strategy from Definition 2.

Suppose, for a contradiction, that  $\Gamma$  is not  $\mathcal{H}^{(r)}(n, 1)$ -universal. Then the embedding strategy must fail at some time  $t^*$  for some  $H$ . This means that there is no available candidate for  $v_{t^*}$ .

Let  $k^*$  be the positive integer chosen by the embedding strategy. We have  $\Delta_{k^*} \leq d_H(v_{t^*}) \leq \Delta_{k^*-1}$  if  $k^* < N$  and  $d_H(v_{t^*}) \leq \Delta_{N-1}$  if  $k^* = N$ , where  $d_H(v_{t^*})$  is the degree of  $v_{t^*}$  in  $H$ .

Let  $B := \{\psi_{t^*-1}(L^-(v_{t^*}))\}$ , notice that  $B$  is empty or contains a single  $(r-1)$ -set. We claim that  $\{B\}$  is a well-behaved multiset. If  $B = \emptyset$ , then  $\{B\}$  is a well-behaved multiset straightforwardly. If  $|B| = 1$ , condition (WB1) is immediate. For (WB2), let  $1 \leq k \leq N$  and let  $u \in W_k$ . Since  $|B| = 1$ ,  $|\{i \in [1] : u \in V(B_i)\}| \leq 1 < 3 \leq \Delta_N \leq \Delta_{k-1}$ , thus (WB2) holds.

For (WB3), let  $1 \leq k \leq N$  and  $1 \leq j \leq \log n$ . Since  $B$  consists of a single  $(r-1)$ -set,  $|V(B) \cap W_{k,j}| \leq r-1$ . On the other hand,

$$|W_{k,j}| \geq \frac{|W_k|}{2 \log n} = \frac{50 \cdot 3^q r n}{\Delta_k \log n} = \frac{50 \cdot 3^q r n^{1-q^{-k}}}{\log n}.$$

Since  $q = r-1 \geq 2$ , we have  $1 - q^{-k} > 0$ . For all sufficiently large  $n$ ,

$$|V(B) \cap W_{k,j}| \leq \frac{1}{2} \frac{50 \cdot 3^q r n^{1-q^{-k}}}{\log n} = \frac{1}{2} |W_{k,j}|$$

. Hence (WB3) also holds. Therefore  $\{B\}$  is well-behaved.

Applying Lemma 2 with the well-behaved multiset  $\{B\}$ ,  $k = k^*$ , and  $j = \log n$ , we obtain

$$\begin{aligned} |\{u \in W_{k^*, \log n} : B \subseteq L(u)\}| &\geq \min \left\{ \frac{1}{16}, \frac{1}{8} p^* \Delta_{k^*} \right\} |W_{k^*, \log n}| \\ &= \min \left\{ \frac{1}{16} \cdot \frac{100 \cdot 3^q r n}{2 \Delta_{k^*} \log n}, \frac{1}{8} \cdot \frac{100 \cdot 3^q r n}{2 \Delta_{k^*} \log n} \cdot \frac{K_r (\log n)^2}{n} \Delta_{k^*} \right\} \\ &= \min \left\{ \frac{100 \cdot 3^q r}{32} \cdot \frac{n^{1-q^{-k^*}}}{\log n}, \frac{50 \cdot 3^q r K_r}{8} \log n \right\} \\ &> 1 \end{aligned}$$

The last inequality uses the facts that  $1 - q^{-k^*} > 0$  and both quantities tend to infinity as  $n$  being sufficiently large. Thus there exists more than one candidate for  $v_{t^*}$  in  $W_{k^*, \log n}$ . Since the embedding strategy fails at time  $t^*$ , every such candidate must belong to  $\text{Im}(\psi_{t^*-1})$ .

Define

$$L_{k,j} := (4 \log n)^{1-j} \begin{cases} \frac{r n}{\Delta_k}, & 1 \leq k \leq N-1, \\ n, & k = N. \end{cases}$$

We claim that for every  $t \leq t^* - 1$ ,  $|\text{Im}(\psi_t) \cap W_{k,j}| \leq L_{k,j}$  for all  $1 \leq k \leq N$  and  $1 \leq j \leq \log n$ .

Assuming this claim is true, we obtain

$$\begin{aligned}
|\text{Im}(\psi_{t^*-1}) \cap W_{k^*, \log n}| &\leq L_{k^*, \log n} \\
&= (4 \log n)^{1-\log n} \frac{rn}{\Delta_{k^*}} \\
&= (4 \log n)^{1-\log n} r n^{1-q^{-k^*}} \\
&= 4r \log n \cdot n^{1-q^{-k^*}-\log(4 \log n)} \\
&= 4r \log n \cdot n^{1-q^{-k^*}-2-\log \log n} \\
&\leq 4r \log n \cdot n^{-1} \\
&\leq 1,
\end{aligned}$$

the last inequality again comes from fact that  $n$  is sufficiently large. If the claim holds, then not all candidates in  $W_{k^*, \log n}$  can be covered by  $\text{Im}(\psi_{t^*-1})$ , contradicting the assumption that the embedding strategy fails at time  $t^*$ .

It therefore remains to prove the claim. We proceed by induction on  $t$ .

For the base case  $t = 1$ , the embedding strategy embeds  $v_1$  into some vertex of  $W_{k,1}$ , since  $L^-(v_1) = \emptyset$ . Hence  $|\text{Im}(\psi_1) \cap W_{k,1}| = 1 \leq L_{k,1}$ , and  $|\text{Im}(\psi_1) \cap W_{k,j}| = 0 < L_{k,j}$  for every  $j > 1$ . Thus the claim holds for  $t = 1$ .

Now let  $1 < t < t^* - 1$ , and assume that  $|\text{Im}(\psi_{t-1}) \cap W_{k,j}| \leq L_{k,j}$  for every  $k$  and  $j$ . We will show that  $|\text{Im}(\psi_t) \cap W_{k,j}| \leq L_{k,j}$  for every  $k$  and  $j$ .

Suppose not. Since the only difference between  $\text{Im}(\psi_t)$  and  $\text{Im}(\psi_{t-1})$  is the newly embedded vertex  $\psi_t(v_t)$ . Therefore there exists a unique pair  $(k', j')$  such that  $\psi_t(v_t) \in W_{k', j'}$  and

$$|\text{Im}(\psi_t) \cap W_{k', j'}| > L_{k', j'}. \quad (\text{A1})$$

For every other pair  $(k, j) \neq (k', j')$ , we still have

$$|\text{Im}(\psi_t) \cap W_{k,j}| \leq L_{k,j}. \quad (\text{A2})$$

We shall show that this is impossible.

**Case 1:**  $j' = 1$ .

If  $k' = N$ , then

$$|\text{Im}(\psi_t) \cap W_{N,1}| > L_{N,1} = n,$$

which is impossible since  $H$  has only  $n$  vertices and  $t \leq n$ .

Now suppose  $k' < N$ . Recall that a vertex of  $H$  is embedded into  $W_{k'}$  only if its degree is greater than  $\Delta_{k'}$ . By Observation 2.1 in [Kat25], the number of such vertices is at most  $\frac{rn}{\Delta_{k'}}$ . Therefore

$$|\text{Im}(\psi_t) \cap W_{k',1}| \leq \frac{rn}{\Delta_{k'}} = L_{k',1},$$

contradicting to assumption A1.

**Case 2:**  $j' > 1$ .

Define

$$\mathcal{B} = \left\{ B_x : B_x = \psi_t(L^-(x)), \psi_t(x) \in W_{k', j'} \right\}.$$

Notice that  $|\mathcal{B}| = |\text{Im}(\psi_t) \cap W_{k', j'}|$ .

We claim that  $\mathcal{B}$  is well-behaved. Condition (WB1) holds since every back-link has size at most 1, and  $\psi_t$  is an embedding.

For (WB2), let  $u \in W_k$ . Suppose  $u \in B_i$  for some  $i$ . Then there exists a vertex  $v_b \in V(H)$  such that  $u \in \psi_t(L^-(v_b))$ . Since  $u \in \text{Im}(\psi_t)$ , there exists a vertex  $v_a$  with  $\psi_t(v_a) = u$ . As  $u$  belongs to the embedded back-link of  $v_b$ , we have  $a < b$ . Therefore

$$|\{i : u \in V(B_i)\}| \leq d_H(v_a) \leq \Delta_{k-1}$$

by the definition of embedding strategy. Hence (WB2) holds.

For (WB3), by the induction hypothesis we have  $|\text{Im}(\psi_{t-1}) \cap W_{k,j}| \leq L_{k,j} + 1$  for every  $k, j$ . We shall show that  $L_{k,j} + 1 \leq \frac{1}{16}|W_{k,j}|$  for every  $k, j$ .

First consider  $j = 1$ . If  $k = N$ , then  $|W_{N,1}| = \frac{1}{2}|W_N| \geq 50rn$ . Since  $L_{N,1} = n$ ,  $L_{N,1} + 1 = n + 1 < \frac{50}{16}rn \leq \frac{1}{16}|W_{N,1}|$  for sufficiently large  $n$ . Now suppose  $k < N$ . Then  $|W_{k,1}| = \frac{1}{2}|W_k| = \frac{50 \cdot 3^q rn}{\Delta_k}$ , and  $L_{k,1} + 1 = \frac{rn}{\Delta_k} + 1 < \frac{50 \cdot 3^q rn}{16\Delta_k} = \frac{1}{16}|W_{k,1}|$  for sufficiently large  $n$ .

Now consider  $j > 1$ . Since  $L_{k,j} = (4 \log n)^{1-j} L_{k,1} = \frac{L_{k,1}}{4 \log n}$ , and  $|W_{k,j}| \geq \frac{|W_{k,1}|}{\log n}$ , we obtain

$$L_{k,j} + 1 \leq \frac{L_{k,1}}{4 \log n} + 1 \leq \frac{|W_{k,1}|}{64 \log n} + 1 \leq \frac{|W_{k,j}|}{64} + 1$$

Furthermore,  $|W_{k,j}| \geq \frac{50 \cdot 3^q rn}{\Delta_k \log n} = \frac{50 \cdot 3^q r n^{1-q-k}}{\log n}$ , which tends to infinity as  $n \rightarrow \infty$ . Hence, for sufficiently large  $n$ ,  $\frac{|W_{k,j}|}{64} + 1 \leq \frac{1}{16}|W_{k,j}|$ . Therefore  $L_{k,j} + 1 \leq \frac{1}{16}|W_{k,j}|$ , and (WB3) follows.

Thus  $\mathcal{B}$  is well-behaved. We may therefore apply Lemma 2 to  $\mathcal{B}$  with  $k = k'$  and  $j = j' - 1$ .

If  $L_{k',j'-1} \geq 1$ , then Lemma 2 gives

$$\begin{aligned} \left| \{u \in W_{k',j'-1} : \exists B \in \mathcal{B}, B \subseteq L(u)\} \right| &\geq \min \left\{ \frac{1}{16}, \frac{1}{8} |\mathcal{B}| p^* \Delta_{k'} \right\} |W_{k',j'-1}| \\ &\geq \min \left\{ \frac{|W_{k',j'-1}|}{16}, \frac{1}{8} |\text{Im}(\psi_t) \cap W_{k',j'}| p^* \Delta_{k'} |W_{k',j'-1}| \right\} \\ &\geq \min \left\{ L_{k',j'-1} + 1, \frac{1}{8} L_{k',j'} p^* \Delta_{k'} |W_{k',j'-1}| \right\} \\ &\geq \min \left\{ L_{k',j'-1} + 1, \frac{1}{8} \frac{L_{k',j'-1}}{4 \log n} \cdot \frac{K_r (\log n)^2}{n} \cdot \Delta_{k'} \cdot \frac{100 \cdot 3^q rn}{2 \Delta_{k'} \log n} \right\} \\ &\geq \min \left\{ L_{k',j'-1} + 1, \frac{100 \cdot 3^q r K_r}{64} L_{k',j'-1} \right\} \end{aligned} \quad (3)$$

As  $L_{k',j'-1} \geq 1$ , either term in equation 3 is at least  $L_{k',j'-1} + 1$ . However assumption A2 assumes the number of candidates cover by  $\mathfrak{S}\phi_t$  is at most  $L_{k',j'-1}$ , which means there is at least one candidate available.

If  $L_{k',j'-1} < 1$ , we have  $|\mathcal{B}| = |\text{Im}(\psi_t) \cap W_{k',j'}| \geq 1$ , as  $\psi_t(v_t) \in W_{k',j'}$ .

Applying Lemma 2 to  $\mathcal{B}$  with  $k = k'$  and  $j = j' - 1$ , we obtain

$$\left| \{u \in W_{k',j'-1} : \exists B \in \mathcal{B}, B \subseteq L(u)\} \right| \geq \min \left\{ \frac{1}{16}, \frac{1}{8} |\mathcal{B}| p^* \Delta_{k'} \right\} |W_{k',j'-1}| \geq \min \left\{ \frac{1}{16}, \frac{1}{8} p^* \Delta_{k'} \right\} |W_{k',j'-1}|$$

We claim that both terms inside the minimum are at least 1:

$$\frac{1}{16} |W_{k',j'-1}| \geq \frac{1}{16} \cdot \frac{50 \cdot 3^q rn}{\Delta_{k'} \log n} = \frac{50 \cdot 3^q r}{16} \cdot \frac{n^{1-q-k'}}{\log n} > 1$$

for sufficiently large  $n$ .

Moreover,

$$\frac{1}{8} p^* \Delta_{k'} |W_{k',j'-1}| \geq \frac{1}{8} \cdot \frac{K_r (\log n)^2}{n} \cdot \Delta_{k'} \cdot \frac{50 \cdot 3^q rn}{\Delta_{k'} \log n} = \frac{50 \cdot 3^q r K_r}{8} \log n > 1$$

for sufficiently large  $n$ .

Therefore  $\left| \{u \in W_{k',j'-1} : \exists B_i \in \mathcal{B}, B_i \subseteq L(u)\} \right| \geq 1$ , which means that there exists a candidate for some  $x$  in  $W_{k',j'-1}$  whose back-link is  $B_i$ . Since  $L_{k',j'-1} < 1$ , the induction hypothesis gives  $|\text{Im}(\psi_{t-1}) \cap W_{k',j'-1}| = 0$ . Thus this candidate is available. This contradicts the definition of  $j'$  as the smallest integer such that  $W_{k',j'}$  contains an available candidate for  $x$ .

This contradiction completes the proof of the claim, and hence the proof of the theorem.  $\square$

## References

[Kat25] Peter Allen Julia Böttcher Jasmin Katz. Bounds for hypergraph universality. 2025.